ITT440 - PARALLEL PROGRAMMING

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1.0 - INTRODUCTION

Target API : https://fakestoreapi.com

Tool: Apache JMeter 5.6.3 - CMD

Tester : AIMAN HAKIM BIN BADRUL ZAMAN (SID : 2024130077)

2.0 - Why Apache JMeter?

- ❖ Jmeter offers a perfect environment for beginners while also providing a large space for skills to grow. It is a low skill floor and high skill ceiling tools.
- ❖ Demonstrating spike, load and soak test easily
- ❖ Free to use app and easy to set up as well as run perfectly on Window

3.0 - Test Environment Setup and Methodology

3,1 Test Environment Setup (visual guide)

For the environment setup, the user can watch the video below.

https://youtu.be/qY2wVylRsRM?si=R5SvOqhR6YLrZSCC

3.2 Methodology

Apache JMeter operates on GUI. Therefore, no codes are required and just need to fill in the settings for the test template.

COMMON SETTINGS	
Server Name	fakestoreapi.com
Protocol	HTTPS
HTTP Method	GET
Path	/products
Content - Type Header	Application / json
Timer	1000 ms (1 seconds)
Listeners	Summary report + Aggregate report

SETTINGS OVERVIEW				
	Load Test	Soak Test	Spike Test	
Thread group name	Load Test	Soak Test	Normal Load	Spike Load
Number of thread	300	150	50	400
Ramp-Up periods (seconds)	60	60	30	5
Duration (seconds)	900 (15 min)	7200 (2 hrs)	600 (10 min)	180 (3 min)
Loop count	Forever			
Scheduler	Enabled			
Target Endpoint	/products			
Timer	1000 ms			
Test File Name	LoadTest.jmx	SoakTest.jmx	SpikeTest.jmx	

4.0 Raw Data Presentation

4.1 Load Test

Performance index

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.994	**500 ms**	**1 sec 500 ms**	**Total**
0.994	500 ms	1 sec 500 ms	HTTP Request

Performance index points at **0.994** which then can be rated at **excellent performance**. This index measures user satisfaction and getting 0.994 means that the user is highly satisfied with the API performance.

Statistics

Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	**209045**	**2**	**0.00%**	**247.69**	**189**	**270013**	**209.00**	**237.00**	**263.00**	**469.00**	**231.75**	**2582.16**	**35.76**
HTTP Request	209045	2	0.00%	247.69	189	270013	209.00	237.00	263.00	469.00	231.75	2582.16	35.76

The Load Test with 300 concurrent users achieved an average response time of 248 ms, with 95% of requests completing under 263 ms. The API processed 231.75 transactions per second with a near-perfect error rate of 0.00%

Load Test Overview

fakestoreapi.com excellently handles load test with near-perfect error rate of 0.00%, proving its reliable performance for free public API.

4.2 Soak Test

Performance Index

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.975	**500 ms**	**1 sec 500 ms**	**Total**
0.975	500 ms	1 sec 500 ms	HTTP Request

An Apdex score of **0.975** is very high. This means users would be **highly satisfied** with the performance even after running continuously for 2 hours.

Statistics

Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	**798152**	**0**	**0.00%**	**346.86**	**235**	**4529**	**330.00**	**426.00**	**473.00**	**713.99**	**110.86**	**1235.19**	**17.32**
HTTP Request	798152	0	0.00%	346.86	235	4529	330.00	426.00	473.00	713.99	110.86	1235.19	17.32

fakestoreapi.com shows a very stable performance with 0 errors throughout 2 hours. The API maintaining a solid average response time of 347 ms and shows a very good endurance for a free public API

Soak Test Overview

The Soak Test with 150 concurrent users ran successfully for 2 hours. The API demonstrated excellent stability with 0% error rate and processed 798,152 requests. The average response time remained consistent at 347 ms, and the Apdex score of 0.975 indicates strong user satisfaction even under sustained load.

4.3 Spike Test

Performance Index

APDEX (Application Performance Index)

Apdex	T (Toleration threshold)	F (Frustration threshold)	Label
0.988	**500 ms**	**1 sec 500 ms**	**Total**
0.988	500 ms	1 sec 500 ms	HTTP Request

Apdex score at 0.988 which shows the API excellent performance in taking sudden burst of users

Statistics

Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label	#Samples	FAIL	Error %	Average	Min	Max	Median	90th pct	95th pct	99th pct	Transactions/s	Received	Sent
Total	**78714**	**0**	**0.00%**	**270.72**	**184**	**21406**	**200.00**	**242.00**	**339.00**	**538.99**	**131.37**	**1463.73**	**20.78**
HTTP Request	78714	0	0.00%	270.72	184	21406	200.00	242.00	339.00	538.99	131.37	1463.73	20.78

The API handled the Spike Test very well, handling a sudden burst up to 400 users with zero errors, an excellent average response time of 271 ms, and strong throughput of 131 requests per second.

Spike Test Overview

The Spike Test with a sudden burst of 400 handled well by the API. It shows that the API is reliable and consistent with an average of 270 ms response time, and the Apdex score of 0.988 indicates that the user is very satisfied with the performance.

5. 0 Conclusion

The fakestoreapi.com demonstrated strong and reliable performance across all three performance tests. It successfully handled a steady load of 300 users, a sudden spike up to 400 concurrent users, and a prolonged endurance test of 150 users for 2 hours with zero errors (0.00% error rate) in all scenarios.

Response times remained consistently good, with average times ranging between 248 ms to 347 ms, and excellent Apdex scores between 0.975 – 0.988, indicating high user satisfaction. The API showed good stability with no major degradation during long-duration testing and effectively managed sudden traffic bursts.